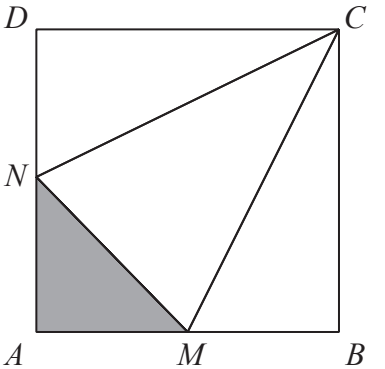
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DO NOT WRITE IN THIS AREA

22 The diagram shows a square *ABCD*.



M is the midpoint of *AB*.
N is the midpoint of *AD*.

The area of the shaded triangle *AMN* is 18 cm^2

Work out the area of triangle *MCN*.

Let AM = AN = t. $\frac{1}{2}t^2 = 18$, so $t^2 = 36$.

1 mark
Use the shaded area.

Square side = $2t$; square area = $4t^2 = 144$.

1 mark
Find whole square.

Area MBC = area NCD = $\frac{1}{2} \times 2t \times t = 36$.

1 mark
Subtract other triangles.

Area MCN = 144 − 18 − 36 − 36 = 54 cm^2

1 mark
54 cm^2.

..... cm^2

(Total for Question 22 is 4 marks)